	B.Tech. Computer Engineering. V Semester II Sessional Test – 2023	
Paper Code: CEN-505	COMPUTER NETWORKS	Max Time: 1 hrs.
	Note: Attempt All Questions	Max. Marks: 15
Q1 CO3	If transmission delay and propagation delay in a sliding window protocol are 1 msec. and 99.5 msec. respectively, then: a. What should be the sender window size to get the maximum efficiency? 200 b. What is the minimum number of bits required in the sequence number field? 8 c. If only 7 bits are reserved for sequence numbers, then what will be the efficiency? 64	5
Q2 CO3	Consider a network connecting two systems located 8000 km apart. The bandwidth of the network is 500×10^6 bits per second. The propagation speed of the media is 4×10^6 meters per second. It is needed to design a Go back N sliding window protocol for this network. The average packet size is 10^7 bits. The network is to be used to its full capacity. Assume that processing delays at nodes are negligible. What is the minimum size in bits of the sequence number field? 8	5
Q3 CO4	Suppose nodes A and B are attached to opposite ends of the cable with propagation delay of 12.5 ms. Both nodes attempt to transmit at t=0. Frames collide and after first collision, A draws k=0 and B draws k=1 in the exponential back off protocol. Ignore the jam signal. At what time (in seconds), is A's packet completely delivered at B if bandwidth of the link is 10 Mbps and packet size is 1000 bits. 37.6	5